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ions. When CO_2 diffuses into the RBCs, it combines with water forming carbonic acid (H_2CO_3) catalyzed by carbonic anhydrase. Carbonic acid is unstable and dissociates into hydrogen and bicarbonate ions.

Carbonic anhydrase facilitates the reaction in both directions.

$$CO_2 + H_2O \underset{\text{carbonic anhydrase}}{\rightleftharpoons} H_2CO_3 \underset{\text{carbonic anhydrase}}{\rightleftharpoons} HCO_3^- + H^+$$

The HCO_3– moves quickly from the RBCs into the plasma, where it is carried to the lungs. At the alveolar site where pCO_2 is low, the reaction is reversed leading to the formation of CO_2 and water. Thus CO_2 trapped as HCO_3^- at the tissue level it is

Events in inspiration and expiration

Inspiration	Expiration
Respiratory centre initiates the stimuli during inspiration.	Respiratory centre terminates the stimuli during expiration.
↓	↓
Impulses are carried to the inspiratory muscles through nerves.	The diaphragm and inspiratory muscles relax.
↓	↓
Diaphragm and inspiratory muscles contract.	Chest wall contracts and the thoracic volume gets reduced.
↓	↓
The thoracic volume increases as the chest wall expands.	The intra pulmonary pressure is reduced.
↓	↓
The intra pulmonary pressure is reduced.	The alveolar pressure increases than the atmospheric pressure.
↓	↓
The alveolar pressure decreases than the atmospheric pressure	Air is sent out due to the contraction of alveoli.
↓	↓
Air flows into the alveoli until the alveolar pressure equalizes the atmospheric pressure and the alveoli get inflated.	Air flows out of the alveoli until the alveolar pressure equalizes the atmospheric pressure and the alveoli get deflated.